Chapter 4
Understanding Generative AI, GenAI

Table of Contents

Chapter 4 1

Understanding Generative AI, GenAI 1

4.1. Introduction 1

4.2. Learning Objectives 2

4.3. Terminology Talk 2

4.4. Additional Resources 7

References 8

4.1. Introduction

Artificial intelligence (AI) is rapidly changing the landscape of education. It can be used to generate engaging learning materials, assist in communication, support personalized learning experiences, and offer real-time feedback on your progress. Generative AI (GenAI) is a powerful tool that's beginning to shape how students learn and interact with content.

But along with these opportunities come important questions—about academic integrity, data privacy, and potential bias in AI systems. As a student, it's essential to understand not just how AI can support your learning, but also the broader implications of its use in education.

In this learning activity, you'll explore how GenAI is being used in educational settings, learn key terms and concepts, and consider both the benefits and challenges of AI in your own learning experience. Whether you're just starting to

learn about AI or have used it before, this activity is designed to help you engage critically and thoughtfully with this evolving technology.

4.2. Learning Objectives

Rationale: Before leveraging AI in your work, it's important to become familiar with the basics of AI. In this Quick Study, you will:

- Explore the context and terminology of GenAI
- Recognize key challenges associated with GenAI in education
- Identify the potential benefits of leveraging GenAI in your work

4.3. Terminology Talk

Artificial intelligence (AI) is a broad field of computer science focused on creating intelligent agents—systems that can reason, learn, solve problems, and act autonomously. While AI systems can learn from the data they're given, they do not possess independent thought or human-like reasoning. Instead, they rely on algorithms and instructions created by humans to define how they process inputs and generate outputs.

One subfield of AI is **generative AI (GenAI)**, which involves developing algorithms that can generate new content such as text, images, music, and more. Examples of GenAI tools include:

- **ChatGPT**
- **Bard**
- **Claude**
- **Copilet**

These tools are trained on massive datasets of existing content and learn to identify patterns and relationships within that data. Once trained, GenAI models can produce new content that is both unique and creative, while resembling the data they were trained on.

Applications of GenAI

GenAI is already being used in various fields, including:

- **Education**: GenAI supports personalized learning by helping educators tailor content to meet individual students' needs.
- **Healthcare**: AI can analyze patient data—including medical history, symptoms, and lab results—to diagnose diseases, predict outcomes, and assist in developing treatment plans.
- **Manufacturing**: AI helps automate repetitive tasks, optimize production processes, and enhance quality control.

As AI continues to evolve rapidly, the possibilities for future applications are vast.

Key Terms to Know

To build a strong foundation in AI, it's important to understand these essential terms:

- **LLM (Large Language Model)**: A powerful AI model trained on vast datasets to generate human-like text and perform natural language tasks such as summarizing, translating, or answering questions.
- **Prompting**: The act of inputting text (a prompt) to guide AI-generated responses. Examples include:
    - *"Write a kid-friendly poem about using manners."*
    - *"Create a travel itinerary for a week-long trip to Italy."*
- **Hallucinations**: Instances where an AI generates incorrect or misleading information. This often happens due to insufficient training data or inherent biases in the data.
- **Bias**: AI models can unintentionally favor certain groups or outcomes due to biased training data.
    - *For example*, if a loan prediction model is trained on a dataset with more information about White applicants than Black applicants, it

may learn patterns biased toward the majority group. This could result in the model unfairly predicting higher default risks for Black applicants—even when they are equally creditworthy.

- Detecting and correcting such biases is essential to ensure fairness and accuracy in AI systems.

Access AI Tools

Below is a suggested list of free AI tools that you may find helpful while practicing your AI skills throughout the Quick Study courses. These are just suggestions, feel free to explore other platforms that better suit your needs or are recommended by your institution.

1. **ChatGPT**
   To access ChatGPT, visit the ChatGPT website and click the **"Log In"** button at the top right of the screen. Enter your username and password in the appropriate fields.
2. **Claude**
   To use Claude, go to the Claude website. Enter the email address you wish to use, then click **"Continue with email."** Check your inbox for a temporary login code and enter it to access the platform.
3. **Gemini**
   To use Gemini, sign in using your **Google Account** at the Gemini website. Access is integrated with your Google credentials.
4. **Bing (Copilot)**
   To use Bing's AI features (formerly known as Bing Chat, now integrated as Copilot), sign in with your **Microsoft account** at bing.com. This account allows you to manage your Microsoft apps, services, and more.
5. **Perplexity**
   To access Perplexity, go to the Perplexity website and click **"Sign In"** to begin. Follow the prompts to log in or create a free account.

You now have a foundational understanding of AI and its terminology. If you're ready to test your knowledge, we invite you to answer the following self-check questions:

Self-check exercise 1

1. What is the term used to describe a sophisticated AI model trained on vast datasets to generate human-like text and excel in natural language tasks?
   a) Large Language Model, LLM
   b) A robot
   c) Natural Language Processor, NLP
2. "Revise the tone of this paragraph to sound like Martin Luther King Jr." is an examples of a _____ in GenAI
   a. Prompt
   b. Dataset
   c. Hallucination
   d. Bias
3. What is the term used to describe when AI produces an incorrect response, due to factors like insufficient data or biases?
   a. Prompt
   b. Hallucination
   c. Bias
   d. Delusion
4. Why might bias be present in a GenAI response?
   a. Biased training data or information has been input into the model
   b. GenAI output does not contain bias.
   c. Human error during the dataset.
   d. GenAI models were not created to ensure.
5. Which of these in NOT the name of a GenAI tool?
   a. Claude
   b. Gen
   c. Gemini
   d. ChatGPT

Class activity 1

This exercise, students will learn how to responsibly use AI by evaluating its responses, citing sources, and understanding its limitations.

Part 1: Ask the AI

1. Research a question or pick a question to ask AI (e.g., *"What were the main causes of World War I?"* or *"How does climate change affect coral reefs?"*).
2. Ask two different AI tools the question and copy the response into a document.

Part 2: Fact Check and Reflect

1. Students must:
    - Compare the two responses from part 1, highlight 3 to 5 key claims or facts from each AI's response, and their similarities and differences between the AI responses.
2. Use credible online sources (encyclopedias, academic websites, news organizations) to verify each claim, from the previous step, and determine which claims make more sense.
3. For each claim, answer:
    - Is this accurate? (Yes/No/Partially)
    - What is your source for verifying it?
    - What did the AI get right or wrong?

Part 3: AI Use Reflection

Discussion or writing prompt:

"How can AI help with learning, and what are the risks of using it blindly?"

Students should reflect on:

- Whether AI made any mistakes.

- If they would rely on AI for schoolwork in the future—and why.
- What ethical concerns come with AI (e.g., plagiarism, bias, misinformation).

4.4. Additional Resources

7 Things You Should Know About Generative AI: This article presented by EDUCASE provides explanations of fundamental AI information: https://er.educause.edu/articles/2023/12/7-things-you-should-know-about-generative-ai.

EDUCAUSE Library provides a wealth of resources, articles, and research on the various aspects of AI's impact and integration in higher education: https://library.educause.edu/topics/infrastructure-and-research-technologies/artificial-intelligence-ai.

"Emerging Technologies and Trends": EDUCASE provides a wide variety of articles and videos on AI: https://er.educause.edu/channels/emerging-technologies-trends

"What is artificial intelligence (AI)?" Resource Page: This resource from IBM provides clear explanations of AI terminology and concepts, suitable for beginners seeking to establish a basic understanding of AI: https://www.ibm.com/topics/artificial-intelligence.

University Center for Teaching and Learning at the University of Pittsburgh offers a collection of resources specifically focused on generative AI and its implications for teaching and learning: https://teaching.pitt.edu/generative-ai-resources-for-faculty/.

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Notes: This chapter content is part of the ACUE Commons training course, *Introduction to Generative AI*.